

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) has emerged as a powerful framework for advancing the reasoning capabilities of large language models. Yet prevailing approaches such as GRPO are systematically undermined by vanishing gradients—a failure mode that we trace to two coupled design choices: the fixed clipping bounds imposed on token-level probability ratios and the step-local standardization of identical rewards, which together produce ineffective gradient updates and leave a large fraction of generated responses unexploited. We introduce **Dynamic Clipping Policy Optimization (DCPO)**, which directly resolves both bottlenecks through two complementary mechanisms: a dynamic clipping strategy that adaptively widens the admissible update interval for each token as a function of its prior probability, thereby restoring token-level exploration, and a smooth advantage standardization procedure that aggregates reward statistics across cumulative training steps to substantially improve the response-level utilization of generated data. DCPO establishes state-of-the-art performance across four benchmarks and four distinct models. On the AIME24 benchmark with the Qwen2.5-Math-7B model, DCPO attains an Avg@1 of 46.7 under greedy decoding and an Avg@32 of 38.8 under 32-fold sampling, decisively surpassing DAPO (36.7/31.6), GRPO (36.7/32.1) and GSPO (40.0/34.9). On the AIME25 benchmark with Qwen2.5-14B, DCPO reaches (23.3/19.0), again outperforming GRPO (13.3/10.5), DAPO (20.0/15.3) and GSPO (16.7/9.9). Beyond accuracy, DCPO improves the nonzero-advantage ratio by an average of 28% over GRPO across the four models, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude compared with both GRPO and DAPO—all while delivering superior end-task performance. Collectively, these results establish that DCPO leverages generated data markedly more efficiently, offering a general and effective foundation for reinforcement learning in large language models.

1 INTRODUCTION

The reasoning capabilities of large language models (LLMs) have become a central focus of contemporary research (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Among the techniques driving this progress, Reinforcement Learning from Verifiable Rewards (RLVR) has established itself as a particularly effective paradigm: by optimizing LLMs against rule-based outcome rewards, RLVR delivers pronounced gains precisely in domains that demand reflective, multi-step decision-making, such as mathematics and coding (DeepSeek-AI et al., 2024).

The GRPO (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) family of algorithms was developed to operationalize RLVR at scale. While the strong empirical performance of GRPO has been validated across numerous studies, recent work such as DAPO (Yu et al., 2025) has exposed two critical and mutually reinforcing limitations. First, GRPO is prone to entropy collapse, in which advantage values decay toward zero during the later stages of training and progressively silence the learning signal. Second, inappropriate weight-clipping discards rarely sampled tokens—precisely the signals that carry the most information—thereby curtailing the diversity of model exploration.

Existing remedies address these symptoms only partially. DAPO, for instance, introduces the Clip-Higher strategy to counter entropy collapse by relaxing the upper clipping bound, together with a Dynamic Sampling strategy that filters out prompts with uniform rewards so that every batch retains only samples carrying valid gradients. These interventions, however, incur two notable and persistent costs. (i) **Response-Level Inefficiency**: dynamic sampling discards and regenerates responses, which

intrinsically lowers sampling efficiency and slows training convergence. (ii) **Token Clipping Deficit**: our analysis establishes that the optimal clipping bound is not constant but varies from token to token, so any uniform upper bound is inherently suboptimal. Even approaches that avoid discarding zero-gradient responses outright leave both limitations fundamentally unresolved.

We introduce **Dynamic Clipping Policy Optimization (DCPO)**, a unified reinforcement-learning method that resolves both limitations at their source through three coordinated design choices. First, we develop a **smooth advantage standardization technique that standardizes advantages across cumulative steps**, directly eliminating the vanishing-gradient problem that arises when responses sharing an identical reward are assigned zero advantage. By aggregating the reward distribution of all cumulatively generated responses with that of the current step, the technique restores useful learning signals to otherwise discarded responses and thereby enables markedly more efficient response-level exploration (see Section 2). Second, we propose a **dynamic clipping mechanism that adaptively adjusts the clipping bounds according to the old token-specific probabilities**, replacing the fixed—whether symmetric or asymmetric—clipping bounds of prior methods. Concretely, tokens assigned lower prior probability are granted correspondingly wider clipping bounds, opening additional room for token-level exploration exactly where it is most informative (see Section 2 and Appendix A.6). Third, DCPO computes the mean loss over the tokens within each individual response rather than over an entire batch; this Only-Token-Mean formulation preserves the relative advantage structure among responses to the same prompt and eliminates the dilution induced by batch-level averaging.

We evaluate these mechanisms on a training corpus that merges Math-DAPO-17k (Yu et al., 2025) with levels 3–5 of the MATH dataset (Liu et al., 2025), assessing performance on four widely used mathematical reasoning benchmarks. Across four distinct models, and using GRPO, DAPO, and—where available—GSPO as baselines, DCPO consistently attains the strongest results, with the largest margins concentrated on the most challenging AIME benchmarks: on AIME24-Avg@32 under 32-fold sampling, DCPO-7B scores 38.8 against 32.1 (GRPO), 31.6 (DAPO), and 34.9 (GSPO); on AIME25-Avg@32, DCPO-14B reaches 19.0 against 10.5 (GRPO), 15.3 (DAPO), and 9.9 (GSPO). These accuracy gains are matched by clear efficiency gains: DCPO raises the average nonzero-advantage ratio by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces token clipping by an order of magnitude compared with all baselines, all while delivering superior end-task performance.

For preliminary background, including the loss and advantage formulations of prior algorithms such as GRPO, DAPO, and GSPO, we refer the reader to Appendix A.2.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Clipping the probability ratio is a standard mechanism in reinforcement learning for stabilizing policy updates and preventing the large, destabilizing parameter shifts that off-policy optimization can induce. Its role becomes especially critical under importance sampling, the setting in which off-policy correction is performed. When importance sampling reweights samples by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$, one obtains an unbiased estimate of $\mathbb{E}_{q(x)}[f(x) \frac{p(x)}{q(x)}]$; the estimate is unbiased, yet its variance can be severely inflated, as characterized in Equation (1).

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right)^2 \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right)^2 p(x) dx \quad (1)$$

To curb this variance inflation, prior works (e.g., Schulman et al. (2017)) constrain the probability ratio within a fixed clipping bound governed by the hyperparameter ϵ , as stated in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

A symmetric fixed clipping bound, however, systematically constrains the model’s exploration of the diversity space, most acutely in regions where the old policy assigns low probability mass. There, the absolute admissible bound on decreasing probabilities is inherently narrow, which suppresses the capacity for meaningful policy updates. Asymmetric clipping bounds, as adopted by DAPO, relax this constraint only partially and leave the underlying limitation intact. GSPO fares no better:

by discarding entire responses whenever the sequence-level importance ratio exceeds its asymmetric bounds, it wastes informative tokens while retaining high-variance ones, a combination that destabilizes training.

Controlling the bias–variance trade-off therefore calls for a more principled alternative—one that makes the constraint depend explicitly on the token probability, as formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3) and enforcing non-negativity of the square-root term by taking its maximum with zero yields a closed-form solution for $r(x)$. The result is a family of **dynamic-adaptive clipping bounds** that vary with the old probability $q(x)$ and the hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

$$0.5 + \frac{1}{2} \sqrt{\max \left(1 - \frac{4\epsilon_{low}}{q(x)}, 0 \right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

This dynamic-adaptive clipping scheme systematically enlarges the admissible policy-update interval in low-probability regions, restoring the exploration capacity that fixed bounds suppress. To guard against the instabilities that arbitrarily wide bounds could introduce, and inspired by the dual-clipping method (Ye et al., 2020), we cap the maximum ratio at 10 for both positive and negative advantages, preventing excessively large importance weights from overtraining the policy. The full derivation of these bounds from the importance-sampling framework, together with a detailed analysis of their behavior, is provided in Appendix A.5 and Appendix A.6.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

In prior methods such as GRPO, DAPO, and GSPO, the advantage $\hat{A}_{new,j}^i$ of the j -th response is obtained by standardizing only the rewards $R_{j=1,\dots,G}^i$ collected for the same prompt at the i -th step, as given in Equation (11) of Appendix A.3. This step-local standardization induces two distinct failure modes. First, whenever sampling randomness drives all rewards at a given step to the same value, the standardized advantage collapses to zero, and the prompt is excluded from parameter updates even though its responses may encode valuable differences in reasoning trajectories. Second, the randomness inherent in high-entropy sampling can yield sharply skewed label counts, producing large step-to-step fluctuations in the standardized advantage—including sign reversals—that destabilize training.

Recent evidence (Yue et al., 2025) indicates that RLVR primarily refines a policy by adjusting the likelihood of correct samples rather than fundamentally reshaping the underlying distribution. This observation licenses a key modeling choice: the reward distribution of responses to a given prompt can be treated as draws from a single global distribution that persists throughout training. That premise directly motivates a *cumulative* standardization, defined in Equation (5).

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the statistics are computed over all responses generated so far for the same prompt.

To temper the residual fluctuations of both the step-specific standardization ($\hat{A}_{new,j}^i$) and the cumulative standardization ($\hat{A}_{total,j}^i$), we introduce two smoothing functions, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$, each formed as a weighted average of the two standardization schemes with step-dependent weights that evolve over the training step i , as defined in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

As training progresses, $\hat{S}A_{new,j}^i$ shifts weight toward the current distribution, encoding a preference for adapting to the latest policy outputs, whereas $\hat{S}A_{total,j}^i$ shifts weight toward the cumulative distribution, treating the current step’s deviation as a fine-grained adjustment within the broader distribution. The final advantage $\hat{A}_j^i$ then selects, between these two smoothed estimates, the one with the smaller absolute value, so as to attenuate whichever source of fluctuation—cumulative or step-specific—is momentarily larger and thereby preserve training stability, as defined in Equation (7).

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

A direct consequence of this construction is that, once a prompt has entered optimization, every response it generates in subsequent steps continues to contribute to the update irrespective of whether its step-local advantage is zero. In particular, when identical rewards occur within a step, the affected responses still update the model through the residual advantage $\frac{1}{i}\hat{A}_{total,j}^i$, so that useful learning signals are retained rather than discarded and data efficiency improves accordingly.

3.1 OTM LOSS

The **Token-Level Mean loss (TLM)** (the loss in Equation (13), used by DAPO) equalizes the contribution of every token, but in doing so it distorts the relative relationship between responses of differing lengths. Consider two responses to the same prompt: response A with advantage 1 and length 500, and response B with advantage 0.5 and length 1500. Under TLM, A is weighted by $\frac{500}{500+1500} = 0.25$ and B by $\frac{1500}{500+1500} = 0.75$, so their total loss contributions become $1 \times 0.25 = 0.25$ and $0.5 \times 0.75 = 0.375$, respectively. The lower-advantage response B thus exerts the greater influence (0.375 versus 0.25), a length-driven inversion of the intended advantage ordering.

The **Sequence-Level Mean loss (SLM)** (the loss in Equation (12), used by GRPO) exhibits a complementary defect: by averaging across the batch it attenuates the magnitude of each advantage, effectively scaling it down by a factor of G . As the batch size grows, this attenuation erodes the correlations that the original advantage standardization was designed to encode. Since standardized advantages already carry well-defined relative relationships across responses, averaging over the $|o_i|$ tokens within a single response both preserves those relationships and distributes importance uniformly across the tokens of that response. We therefore average only over the tokens within each response and deliberately omit the batch-level average across responses—a formulation we term the **Only Token Mean loss (OTM)**, given in Equation (8).

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}} \right) \hat{A}_{i,t} \right) \quad (8)$$

Following DAPO, the DCPO loss omits the KL-divergence constraint, and its lower and upper clipping bounds are instantiated by the dynamic-adaptive bounds introduced in Section 2.

4 EXPERIMENTAL SETUP

Models and Datasets To probe generality across both model scale and model type, we experiment with four models spanning instruct and base variants: Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b). Training draws on a merged corpus of approximately 25k mathematical problems that combines the DAPO-Math-17K corpus (Yu et al., 2025) with levels 3–5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). Every problem is formatted with the Qwen-Math template (see Appendix A.8) for both training and evaluation, ensuring a consistent prompting protocol throughout.

Evaluation We evaluate on four widely used mathematical reasoning benchmarks that together span a broad difficulty spectrum: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). For **MATH500**, we report Avg@1 alone,

as its larger size (500 problems) already yields a stable estimate. For the smaller **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems) benchmarks, we report both Avg@1 and Avg@32, so that greedy and sampling-based behavior are each assessed under a robust and comprehensive protocol.

- **Avg@1**: the standard accuracy under greedy decoding, quantifying the quality of the model’s single most probable prediction.
- **Avg@32**: the mean accuracy over 32 responses sampled per problem at temperature 1.0 and top_p 1.0, quantifying the robustness and stability of the trained policy distribution under stochastic decoding.

Baseline Settings We adopt GRPO and DAPO as baselines throughout, and additionally include GSPO on Qwen2.5-Math-7B and Qwen2.5-14B. The remaining configuration—including the per-method clipping thresholds ϵ_* , the codebase, and the common training hyperparameters—is detailed in Appendix A.9.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1 reports the performance of GRPO, DAPO, GSPO, and our proposed DCPO on the four mathematical reasoning benchmarks—MATH500, AMC23, AIME24, and AIME25—under both greedy decoding (Avg@1) and sampling-based decoding (Avg@32), together characterizing each method’s robustness and generalization. Across every model scale, from 1.5B to 14B, and across both instruct and base variants, DCPO attains the strongest average performance, matching or surpassing GRPO, DAPO, and—where available—GSPO. This uniformity across model capacities and decoding regimes establishes DCPO as a general-purpose policy optimization method for reinforcement learning of LLMs rather than a benchmark-specific tuning.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO achieves the highest average scores (47.2/32.8), exceeding GRPO (+1.2/+0.8) and DAPO (+0.7/+0.4). On Qwen2.5-3B, DCPO reaches (37.6/22.7), surpassing GRPO (+1.3/+1.7) and remaining competitive with DAPO—improving Avg@1 by +1.0 while trailing by a marginal -0.4 on Avg@32. On Qwen2.5-Math-7B, DCPO attains (57.1/45.2), outperforming GRPO (+4.0/+3.6), DAPO (+3.3/+2.8), and GSPO (+1.9/+1.9). On the largest model tested, Qwen2.5-14B, DCPO delivers (53.2/39.0), with substantial gains over GRPO (+6.5/+7.7) and consistent advantages over DAPO (+1.3/+0.1) and GSPO (+4.2/+5.5) in Avg@1 and Avg@32. Taken together, these results demonstrate that DCPO improves reliably across a wide range of model capacities and aligns the policy more effectively to the reward signal under both greedy and stochastic decoding.

Robustness on AIME under Sampling (Avg@32) The most pronounced advantages emerge on AIME24 and AIME25, the high-difficulty competition benchmarks that most severely stress a model’s reasoning ability. Under Avg@32—which measures the capacity to produce correct responses under high-entropy sampling—DCPO exhibits markedly stronger robustness at the 7B and 14B scales: on AIME24-Avg@32, DCPO-7B achieves 38.8, ahead of GRPO (32.1), DAPO (31.6), and GSPO (34.9); on AIME25-Avg@32, DCPO-14B reaches 19.0, ahead of GRPO (10.5), DAPO (15.3), and GSPO (9.9). Because the sampling metric rewards the diversity and reliability of correct generations rather than a single greedy output, these margins indicate that DCPO improves not only average-case accuracy but also the breadth of correct reasoning trajectories the policy can realize. This joint optimization of reward fidelity and exploration under high-entropy decoding is precisely the property demanded by deployment settings in which many plausible outputs are sampled.

5.2 THE TOKEN CLIPPING RATIO (*TCR*)

We define the **Token Clipping Ratio** (*TCR*) as the proportion of tokens that clipping excludes from the policy-gradient update during back-propagation; a lower *TCR* therefore reflects a larger fraction of the generated sequence contributing to learning. As Figure 1 shows, the *TCR* behaves in sharply different ways across reinforcement-learning methods.

$$TCR = Average \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } micro_m}{\text{Total number of tokens in } micro_m} \right) \quad (9)$$

Table 1: Performance across the four mathematical reasoning benchmarks, reported per model. Avg@1 uses greedy decoding; Avg@32 averages over 32 sampled responses. Boldface marks the best value in each column, and Δ_{baseline} reports DCPO’s margin over the corresponding baseline (red indicates an improvement).

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/ 70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

where $micro_m$ denotes the m -th microbatch and N is the number of microbatches per training step.

Under GRPO, the TCR follows model-size-dependent trajectories: it rises with training steps for the smaller models (Qwen2.5-Math-1.5B and Qwen2.5-3B) yet gradually declines for the larger ones (Qwen2.5-Math-7B and Qwen2.5-14B). DAPO, by contrast, exhibits a monotonically increasing TCR across all models, meaning that as training proceeds an ever-larger share of policy updates is computed from partial or truncated responses. Both GRPO and DAPO, moreover, display pronounced fluctuations and occasional anomalous spikes in the TCR , which inject instability into optimization and can precipitate entropy collapse. GSPO is more wasteful still: its average TCR exceeds 11% on Qwen2.5-Math-7B and 15% on Qwen2.5-14B—far above that of token-level clipping methods—and discards a correspondingly larger volume of informative tokens.

DCPO, in stark contrast, sustains a stable and efficient training regime. Independent of model size or training stage, its TCR stays nearly constant and remains an order of magnitude below that of GRPO and DAPO. This uniformly low and stable TCR means that DCPO discards far fewer tokens, allowing a greater share of complete response sequences to inform each policy update and thereby opening additional room for the model to explore rare-token diversity. The result is a clear improvement in sample efficiency over the baselines, achieved while keeping outlier variation tightly bounded.

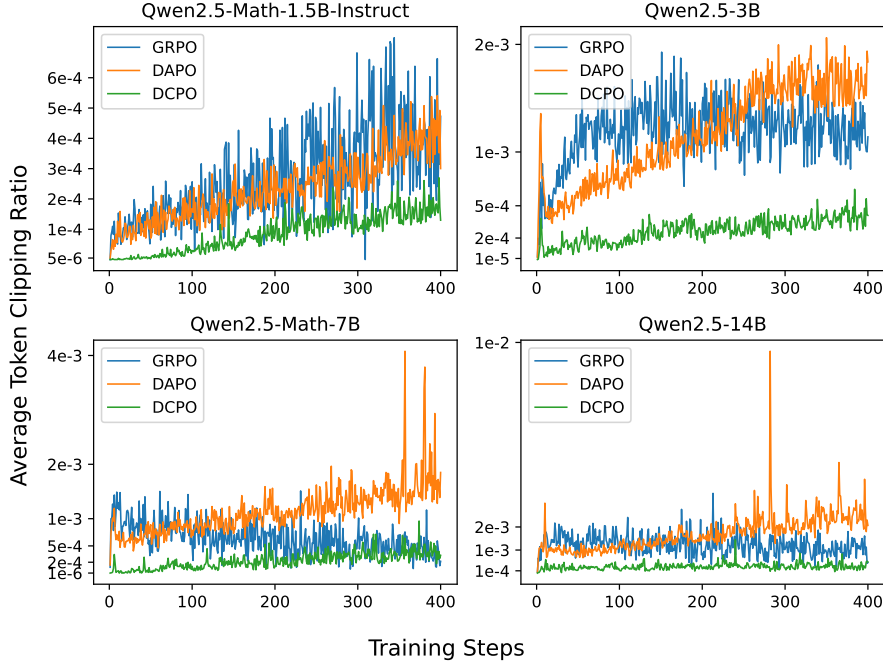


Figure 1: Token Clipping Ratio (*TCR*) over training for each method across the four models. DCPO sustains a substantially lower and more stable *TCR* than GRPO and DAPO throughout training.

Taking Qwen2.5-Math-7B as a representative case, the token distribution concentrates rapidly on high-confidence tokens: after roughly 60 steps about 95% of generated tokens satisfy $q(x) > 0.9$, a fraction that rises to about 97% by roughly 100 steps and keeps climbing thereafter. As illustrated in Figure 4d, once the old probability satisfies $q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83$, the new probability $p(x)$ contributes to the update within the interval $[\frac{1}{1+\epsilon}, 1]$ and is never discarded under any clipping method. The overwhelming majority of tokens therefore reside in a region that is inherently immune to clipping, which explains the low overall *TCR* observed in practice. Conversely, Figure 1 shows the *TCR* rising during the later phases of most runs, signaling that a growing proportion of the remaining low-probability tokens is being discarded—precisely the tokens whose loss most restricts the model’s ability to explore rare-token diversity.

Prior work (Wang et al., 2025) establishes that **high-entropy tokens**—which are exactly the low-probability tokens—are the primary drivers behind the emergence of reasoning capabilities in LLMs. Since entropy and probability are inversely related, rarer tokens carry proportionally more information; granting such tokens a wider admissible update range enlarges the model’s effective exploration space and directly facilitates the acquisition of diverse reasoning behaviors.

These empirical and theoretical observations jointly motivate our **dynamic-adaptive clipping** scheme, which relaxes the clipping constraint for tokens with $q(x) < \frac{1}{1+\epsilon}$ while retaining the safe interval for high-confidence tokens. By adaptively widening the update interval for the high-entropy tail of the distribution, the method restores the model’s capacity to learn from rare, informative tokens and consequently strengthens its reasoning ability.

5.3 THE RESPONSE UTILIZATION RATIO (*RUR*)

During RLVR training, a response contributes to a policy update only when it carries a nonzero gradient. We quantify how effectively this condition is met with the **Response Utilization Ratio** (*RUR*), the percentage of generated responses with nonzero advantage that actually participate in the policy update.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

As Table 2 summarizes, the average *RUR* under GRPO is strikingly low—only 44.6% averaged across the four models—so that more than half of all generated responses are discarded, a substantial inefficiency in exploiting the diversity of exploration. DCPO, by contrast, sustains an average *RUR* of roughly 70% after the first epoch, an absolute gain of 28% and a relative improvement of 64% over GRPO.

Table 2: Average *RUR* over 400 training steps, per model. DCPO raises response utilization to roughly 70% across all models, well above GRPO and GSPO (“-” denotes settings in which GSPO was not evaluated).

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 shows that GRPO’s *RUR* degrades sharply over the course of training, independent of model size or type, falling from above 90% to below 50%; on Qwen2.5-Math-7B it collapses to as low as 30% after roughly 200 steps. This decline is the direct signature of GRPO’s step-local advantage standardization, which produces a large and growing fraction of zero gradients and thereby renders exploration increasingly inefficient in the later stages of training. DCPO, driven by smooth advantage standardization, instead maintains a consistently high *RUR*: after the first epoch it stabilizes around 70% and continues to rise with further training.

The *RUR* exceeds 90% during the first few steps but, from the second epoch to the end of training, settles around 70% rather than reaching 100%. Examining why illuminates the value of the SAS mechanism.

First, the initial *RUR* above 90% is largely an artifact of early-stage outputs that violate the required formatting, incur negative rewards, and thereby inflate the metric. That these outputs are quickly corrected shows the models rapidly internalize fixed answer patterns, which is why subsequent training should shift emphasis toward exploring genuine reasoning capabilities. Second, our analysis identifies two principal sources of persistently zero advantage that pin the *RUR* near 70% rather than at 100%. (i) Nearly half of the problems are easy enough that the model solves them consistently; excluding these from updates is justified, since the model’s core capabilities require no further refinement on them. (ii) The remaining consistently zero-advantage cases correspond to persistent errors, which arise for two reasons. On the one hand, some problems remain too difficult for the current model even given a correct ground-truth label. On the other hand, the DAPO-Math-17K dataset contains items whose labels were converted to mislabeled integers, introducing spurious “incorrect” targets. In both cases, discarding zero-advantage responses functions as an implicit filter, mitigating the risk of updating the model on noisy or mislabeled data.

Because DAPO, unlike GRPO and DCPO, discards responses with identical labels from the generated data, a direct *RUR* comparison is not meaningful; we therefore gauge DAPO’s efficiency through the number of generated responses and the wall-clock time required to complete an equal number of update steps. To offset the low sampling efficiency that GRPO’s low *RUR* would otherwise cause, we set DAPO’s generation batch size to 1–4 times the training batch size and adjust it dynamically. Across the four models, DAPO nonetheless requires 3 to 5 times more generated responses than DCPO to reach the same number of parameter updates, at least doubling the training GPU hours while yielding no performance gain. This inefficiency worsens with scale: on Qwen2.5-14B, the fraction of responses DAPO actually uses falls below 30% after about 300 steps. Dynamic sampling thus combines low data utilization with poor training efficiency and forfeits substantial opportunity to explore model diversity.

In summary, DCPO advances decisively beyond both GRPO and DAPO: it substantially raises response utilization and training efficiency, converts the full pool of sampled responses into usable exploration signal for model diversity, and translates that efficiency into improved end-task performance.

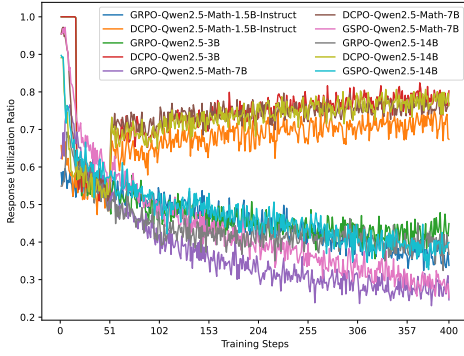


Figure 2: Evolution of the *RUR* during training. GRPO decays sharply while DCPO stabilizes near 70%.

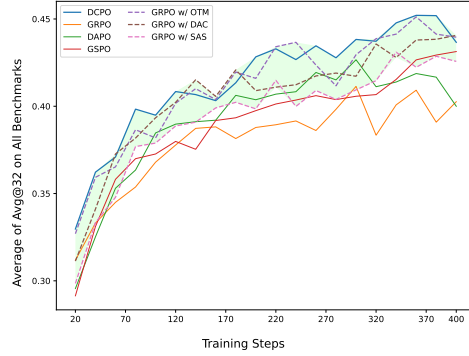


Figure 3: Component ablation on Qwen2.5-Math-7B, measured by average Avg@32 over training. Each DCPO component (OTM, SAS, DAC) is added individually to GRPO, and their combination yields the full DCPO.

5.4 ABLATION STUDIES

To isolate the contribution of each DCPO component, we conduct an ablation on Qwen2.5-Math-7B using **Avg@32**, the metric most sensitive to the robustness and stability of the learned policy distribution. For a controlled comparison, each variant modifies exactly one component of the baseline GRPO framework while holding all other settings fixed and removing the KL-divergence term, so that the configuration aligns with DAPO, GSPO, and the full DCPO. The corresponding learning curves from step 20 to step 400 are shown in Figure 3.

Each of the three components, added in isolation, drives a continuous improvement in reasoning ability: every variant surpasses GRPO and matches or exceeds DAPO and GSPO. Their combination then produces a substantial aggregate gain over all baselines, confirming the effectiveness of the full DCPO pipeline for reinforcement-learning tasks. The distinction sharpens in the later stages of training, where the baselines plateau while DCPO continues to trend upward, both at the level of individual components and in aggregate. The ablation thus establishes that each component contributes positively to training efficiency and that their integration yields cumulative gains and greater stability. We detail each contribution below.

(i) **Only-Token-Mean loss (OTM)**: aggregating the loss within each individual response rather than across the whole batch already lifts GRPO above the baselines, which suffer from lower data utilization and more volatile performance. (ii) **Smooth Advantage Standardization (SAS)**: replacing the stepwise advantage with the cumulative smoothed advantage markedly improves GRPO and approaches DAPO, evidence that aggregating reward statistics across training steps stabilizes optimization and raises final performance. (iii) **Dynamic Adaptive Clipping (DAC)**: substituting a probability-adaptive clipping boundary for GRPO’s fixed boundary yields the largest single-component gain—DAC alone surpasses GRPO, DAPO, and GSPO, and does so more stably than OTM. (iv) **Full DCPO (OTM + SAS + DAC)**: integrating the three mechanisms delivers the best and most stable results overall, decisively outperforming GRPO, DAPO, and GSPO. Their synergy confirms that each design choice contributes a distinct benefit to data-efficient, stable reinforcement learning, jointly improving both performance and robustness for large language models.

Overall, the ablation confirms that every DCPO component contributes positively to end performance and that their combination produces substantial cumulative gains, validating the proposed mechanisms as effective levers for improving data efficiency and stability in reinforcement learning for LLMs.

6 CONCLUSION

This work resolves two key limitations of existing policy optimization methods: the restricted token-level exploration imposed by fixed clipping in low-probability regions, and the low sample efficiency induced by step-local reward standardization. We introduced **Dynamic Clipping Policy Optimization (DCPO)**, a reinforcement-learning pipeline that strengthens the reasoning capabilities

of large language models through two complementary innovations: (i) a **dynamically adaptive token-level clipping** mechanism that widens the update interval as a function of the old-policy probability, enabling efficient exploration of rare, low-probability tokens, and (ii) a **Cumulative-Smooth Advantage Standardization** that aggregates reward statistics across optimization steps, damping the reward randomness introduced by high-entropy sampling. Extensive experiments on four mathematical reasoning benchmarks across four models substantiate these claims: DCPO-7B, for instance, attains an Avg@32 of 38.8 on the challenging AIME24 benchmark, surpassing GRPO (32.1) and DAPO (31.6). Beyond accuracy, DCPO raises the response utilization ratio by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude. Together these results establish DCPO as a method that markedly improves data utilization and diversity exploration in RLVR. Future work will extend DCPO to further domains, such as code generation and semantic reasoning.

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A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

In prior methods, including GRPO and DAPO, the advantage $\hat{A}_{j,t}^i$ for token t in response j is obtained by standardizing the reward R_j^i against the mean μ^i and standard deviation σ^i of the rewards over the G responses generated at the i -th step, as shown in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

When the rewards for a given prompt are identical, the corresponding responses receive zero advantage and contribute nothing to the update, wasting the sampled data.

A.3.1 GRPO

Shao et al. (2024) proposed **Group Relative Policy Optimization (GRPO)**, which samples G responses per query, assigns rewards R via a rule-based reward function, and estimates token-level advantages as in Equation (11). The model parameters are then updated with Equation (12), the **Sequence-Level Mean loss (SLM)**:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level advantages over all tokens in a response and then over all G responses in the batch; the $\mathbb{D}_{\text{KL}}$ penalty constrains the policy against excessive deviation from a reference policy (e.g., a pre-trained model); and the fixed clipping threshold ε bounds the change in token probability ratios to prevent overly large updates. Widely adopted as it is, this batch-level averaging inherently dilutes per-response importance.

A.3.2 DAPO

Building on GRPO, Yu et al. (2025) propose **Dynamic sAmpling Policy Optimization (DAPO)**. Observing that longer sequences exert a disproportionate influence on the GRPO gradient, they normalize by the total token count across the entire batch, yielding the **Token-Level Mean loss (TLM)** in Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. DAPO discards every prompt whose responses all share an identical reward and regenerates responses to restore the batch size, a procedure that incurs severe data inefficiency.

A.3.3 GSPO

Zheng et al. (2025) propose **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping injects high-variance noise into gradient estimates—noise that accumulates over long sequences and is further amplified by the clipping operation itself. GSPO mitigates this by replacing token-wise clipping with sequence-wise clipping, producing a more stable training signal.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

$$\text{where } s_i(\theta) = \exp \left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log \left(\frac{\pi_\theta(o_{j,t}|q)}{\pi_{\theta_{\text{old}}}(o_{j,t}|q)} \right) \right)$$

This sequence-level criterion, however, discards more than 10% of responses that carry nonzero advantages, so its stability comes at the price of considerable data waste and reduced training efficiency. And while GRPO discards responses with sequence-level high variance, it still retains a large number of responses with token-level high entropy, which continue to destabilize training.

A.4 REWARD CALCULATION

To supply a verifiable and interpretable supervision signal, we adopt a rule-based reward model that scores each generated response jointly on answer correctness and format compliance, as defined in Equation (15). Concretely, a reward of +1 is assigned when both the format and the answer are correct, 0 when the format is correct but the answer is incorrect, and −1 when the format is incorrect.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j indexes the response within the group for a given problem at the i -th training step, and the reward is determined jointly by the format and correctness of that response.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Clipping the probability ratio is a crucial technique in reinforcement learning for stabilizing policy optimization and preventing large, destabilizing parameter updates. Its importance is heightened whenever importance sampling is used to estimate expected values under a new policy from samples drawn by an old policy.

The expected value of a function $f(x)$ under the new probability $p(x)$ can be re-expressed as an expectation under the old probability $q(x)$ via the importance-sampling weight, as shown in equation 16:

$$\int f(x)p(x) dx = \int f(x)\frac{p(x)}{q(x)}q(x) dx = \mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \quad (16)$$

Although this estimator is unbiased, its variance can be substantially inflated—a well-known difficulty of importance sampling. The true variance of $f(x)$ under the new policy is given by Equation (17), whereas the variance of the importance-sampled estimator under the old policy is given by Equation (18):

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p}[f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x)\frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The difference between the importance-sampled variance and the true variance makes the potential for inflation explicit, as shown in Equation (19):

$$\text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To contain this discrepancy, many prior works—including PPO Schulman et al. (2017) and RLVR methods (DeepSeek-AI et al., 2024; Liu et al., 2025)—impose a fixed symmetric clipping bound on the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

A fixed clipping bound, however, inherently constrains the policy’s capacity for exploration, a limitation that is most damaging where the old policy assigns low probability mass. In such regions the absolute bound on low probabilities is severely restricted, suppressing the potential for meaningful policy updates. Asymmetric clipping bounds, as introduced by DAPO, address only the symmetry of the constraint; the fundamental limitation of a non-adaptive bound persists.

Moreover, recent studies such as Yue et al. (2025) indicate that the reasoning paths exploited by RLVR models already reside within the sampling distribution of the base model—a conclusion reinforced by the minimal parameter shift observed during off-policy batch updates, which manifests as a small fraction of clipped parameters. Motivated by this stability, we propose a more principled alternative that constrains the probability ratio $r(x)$ through a dynamic-adaptive mechanism, incorporating the probability directly into the restriction as in Equation (21) and thereby replacing the conventional fixed clipping of Equation (20).

$$\left| \left(\frac{p(x)}{q(x)} - 1 \right) p(x) \right| \leq \epsilon \quad (21)$$

Substituting $p(x) = r(x)q(x)$ into the preceding inequality and expanding yields the expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For this clipping ratio to be mathematically valid and meaningful, it must respect the fundamental constraints on probability distributions and their ratios. As stated in Equation (23), the probabilities $p(x)$ and $q(x)$ lie in $[0, 1]$; consequently the probability ratio $r(x)$ is non-negative, as is the clipping hyperparameter ϵ .

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To guarantee the validity and practical applicability of the clipping bound derived from Equation (22), we specify the necessary conditions on its bounds, as set out in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

Combining Equation (22) with Equation (24), the probability ratio $r(x)$ must satisfy the following conditions:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

For the clipping bound, we restrict attention to the overlapping region centered on the ratio value 1, which minimizes the discrepancy between the two conditions and preserves consistency with the conventional fixed bound of GRPO; the resulting validity conditions are:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, we guarantee that the radicand remains non-negative by taking its maximum with zero, e.g. $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. This finally yields the following conditions, which constitute the **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2}\sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower bounds in Inequality 27, we assign distinct thresholds ϵ_{high} and ϵ_{low} so that the dynamic-adaptive bound coincides with the GRPO bound at specific reference points.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To make our dynamic-adaptive clipping bounds coincide with the fixed symmetric bounds at the boundary points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as shown in Figure 4d, we derive the specific values of ϵ_{low} and ϵ_{high} by solving the equations in Equation (28).

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \end{cases} \quad (28)$$

$$\begin{cases} q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases}$$

Under the standard practice of prior work (e.g., GRPO), where the fixed clipping threshold is $\epsilon = 0.2$, solving these equations gives our hyperparameters as approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. Although $\epsilon_{low} \neq \epsilon_{high}$, this asymmetry serves a fundamentally different purpose from the asymmetric fixed clipping of DAPO: here it calibrates a probability-dependent bound to the fixed bound at the reference points rather than statically relaxing the upper limit.

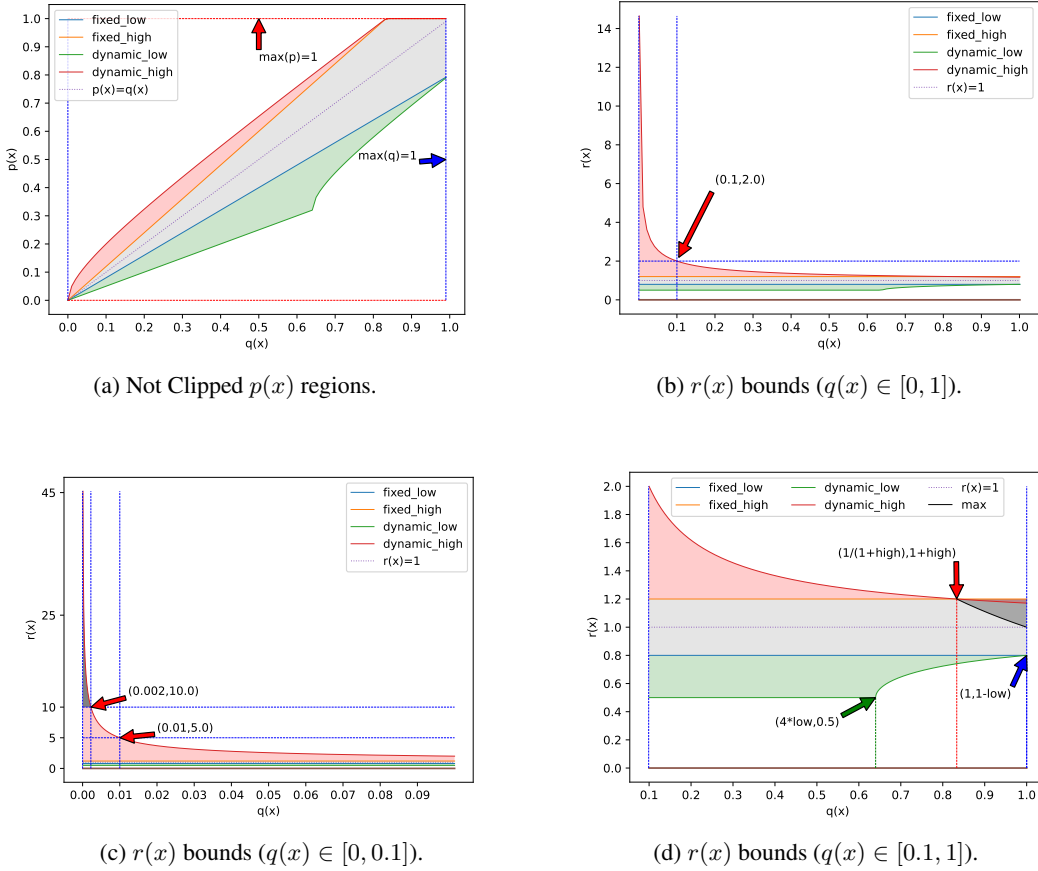


Figure 4: Comparison of clipping bounds as a function of the old probability $q(x)$: fixed clipping ($\epsilon = 0.2$) versus dynamic-adaptive clipping ($\epsilon_{low} = 0.16, \epsilon_{high} = 0.2$). The dynamic bound widens as $q(x)$ decreases, expanding the admissible update interval for rare tokens while coinciding with the fixed bound for high-probability tokens.

Dynamic versus fixed clipping: Figure 4 contrasts the behavior of the fixed clipping scheme used in GRPO ($[1 - \epsilon, 1 + \epsilon]$) with that of our probability-aware adaptive clipping. For a token with old probability $q(x)$, the fixed scheme supplies a constant relative width 2ϵ ; as $q(x)$ decreases, the absolute exploration space it permits shrinks linearly (Figure 4a). The dynamic scheme, by contrast, expands its relative width inversely with $q(x)$, so the admissible exploration space stays approximately constant across the entire probability range.

Effect of decreasing $q(x)$: As $q(x)$ becomes small, the fixed interval $[1 - \epsilon, 1 + \epsilon]$ remains independent of $q(x)$, so the model gains no additional room to explore low-probability tokens; the

dynamic-adaptive bound, in contrast, expands rapidly (Figure 4b), granting a larger relative space to rare tokens—precisely the behavior required for efficient RL.

Behavior for extremely low probabilities: Figure 4c depicts the clipping interval for $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the dynamic-adaptive upper bound on $r(x)$ grows rapidly, at a rate inversely proportional to the square root of $q(x)$, whereas the fixed bound stays constant. Inspired by the dual-clipping strategy (Ye et al., 2020), we impose a hard ceiling $r_{\max} = 10$ to forestall the attendant instabilities (gradient vanishing, explosion, etc.). The gray region in Figure 4c, corresponding to approximately $q(x) \leq 2 \times 10^{-3}$, is where the adaptive upper limit is clamped to $r_{\max}$.

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\epsilon$ the dynamic-adaptive scheme yields $dynamic_{low} = 0.5$, whereas the fixed scheme retains the constant $fixed_{low} = 1 - \epsilon = 0.8$; once $q(x)$ exceeds this threshold, the dynamic-adaptive lower bound contracts smoothly toward the fixed value. Symmetrically, the dynamic-adaptive upper bound exceeds the fixed bound whenever $q(x) < \frac{1}{1+\epsilon_{high}}$ and merges with it for larger $q(x)$. The adaptive interval $[dynamic_{low}(q(x), \epsilon_{low}), dynamic_{high}(q(x), \epsilon_{high})]$ therefore delivers a probability-dependent widening of the admissible region for rare tokens while preserving the tightness of the fixed interval for common tokens (Figure 4d).

Positive versus negative advantage: The loss in Equation (8) comprises both a positive- and a negative-advantage term. Under the minimum operation, the ratio $r(x)$ for a positive advantage is clipped to the interval $[0, dynamic_{high}(q(x), \epsilon_{high})]$, whereas for a negative advantage it is clipped to $[dynamic_{low}(q(x), \epsilon_{low}), \infty)$. Because an excessively large $r(x)$ can destabilize training, we cap the ratio at $r_{\max} = 10$ for both cases; this safeguard, following Ye et al. (2020), ensures that the adaptive limits never produce excessively high importance weights.

In short, for both positive and negative advantages, dynamic-adaptive clipping affords the model more room to explore diversity when rare tokens carry low probability than fixed clipping does.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i index the iteration at which prompts are used to generate responses and update the model parameters. We define the following statistics for reward normalization.

The mean rewards of the responses are given by the group of functions in Equation (29):

$$\begin{cases} \mu_{new}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i = \frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total} = \frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{cases} \quad (29)$$

The corresponding variances of the response rewards are given by the group of functions in Equation (30):

$$\begin{cases}
\sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\
\sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{old}^k)^2} \\
\sigma_{total}^i = \sqrt{\frac{1}{G * i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^k)^2} \\
= \sqrt{\frac{1}{i} \left(\sigma_{new}^i{}^2 + (i-1) \sigma_{old}^i{}^2 + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)}
\end{cases} \quad (30)$$

To reduce resource usage, we compute σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29) through the equivalent recursions in terms of μ_{new}^i , σ_{new}^i , μ_{old}^i , and σ_{old}^i , rather than by direct accumulation over the original rewards.

A.8 THE TEMPLATE FOR EXPERIMENTS

All experiments in this paper use the following prompting template, adapted from the Qwen-Math template (Yang et al., 2024b).

Qwen-Math Template

```

<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant

```

A.9 TRAINING SETTING

We configure the clipping thresholds per method as follows: a symmetric threshold $\epsilon = 0.2$ for GRPO; lower/upper thresholds $(0.2, 0.28)$ for DAPO, augmented with a 512-token soft-punish cache and a maximum probability ratio of 10 for negative advantage; lower/upper thresholds $(3e-4, 4e-4)$ for GSPO; and low/high thresholds $(0.16, 0.2)$ for DCPO. The DCPO thresholds are chosen to preserve the same clipping points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1+\epsilon)$ and $(q(x), r(x)) = (1, 1-\epsilon)$ as GRPO (Figure 4d); unlike DAPO, we cap the maximum probability ratio at 10 for both positive and negative advantages.

Training runs for 400 steps with a generation batch size of 512 and a parameter-update mini-batch size of 32. Using temperature 1 and top-p 1, we generate $G=16$ responses per problem, with the prompt and response maximum lengths set to 1,024 and 3,072 tokens, respectively—the latter far below the 20k maximum used in the original DAPO.

Our training codebase is adapted from Verl (Sheng et al., 2024) and runs on $32 \times H20$ GPUs; unless stated otherwise, hyperparameters match those of DAPO. For loss computation we retain the exact $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted-average formulation of GRPO and DAPO rather than the approximate weighting used in Verl, ensuring correctness across different microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

Avg@1 Trend Figure 5 reports the average Avg@1 metric, obtained by averaging the greedy-decoding (Avg@1) results over the MATH500, AMC23, AIME24, and AIME25 benchmarks, which characterizes the model’s best-prediction performance. Across all four models, DCPO consistently outperforms GRPO and DAPO, and it also surpasses GSPO on the two models where GSPO

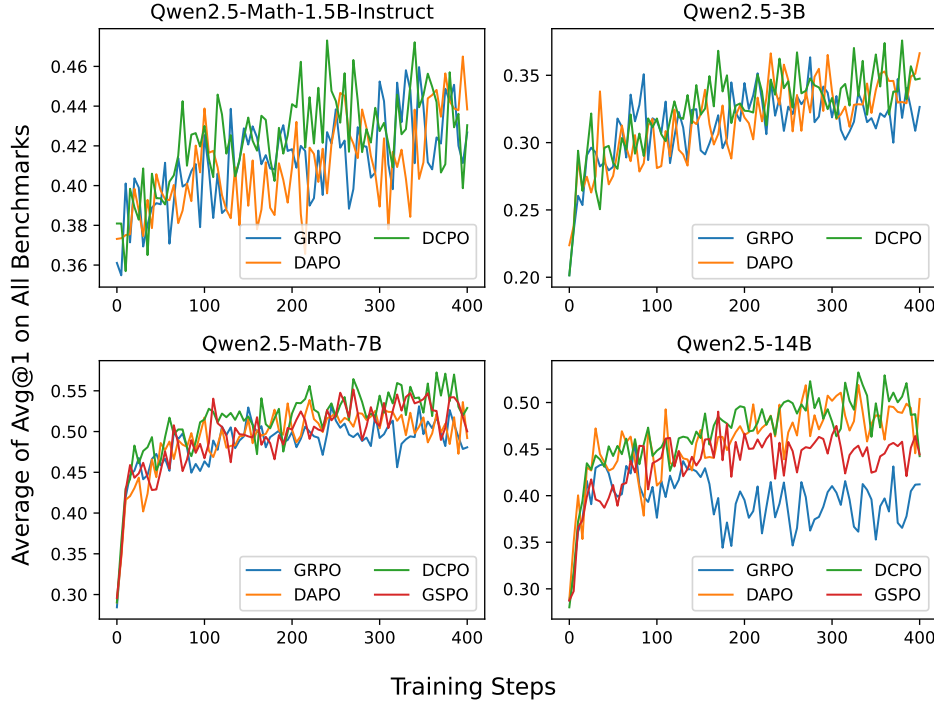


Figure 5: Avg@1 performance over training, averaged across the four benchmarks, for each model and method.

was evaluated. Its trajectory is moreover more stable than the baselines’, free of the pronounced performance drop exhibited by GRPO and the large fluctuations exhibited by DAPO. These trends indicate that DCPO elicits stronger reasoning capabilities during training and leaves greater headroom for the model to reach still higher performance.

Avg@32 Trend Figure 6 compares the Avg@32 performance of DCPO against GRPO, DAPO, and—where available—GSPO across model sizes on the AMC23, AIME24, and AIME25 benchmarks, and demonstrates DCPO’s exceptional robustness and stability under sampling. GRPO, by contrast, exhibits substantial fluctuation and occasional collapse—on Qwen2.5-14B it suffers a marked performance drop after approximately 170 steps. Across every evaluated model (1.5B to 14B), the DCPO trajectory trends consistently upward with minimal fluctuation, sustaining the fastest early-stage growth rate and preserving that upward trend thereafter. Relative to DAPO, DCPO furthermore attains superior performance while halving training time and resource consumption. These results establish the effectiveness and robustness of dynamic-adaptive clipping and smooth advantage standardization in improving exploration of diversity at both the response and label levels.

Entropy Trend Figure 7 shows that, after an initial warm-up, GRPO’s entropy decays sharply and then settles into a flat, low-entropy regime. This entropy collapse reflects a rapid loss of policy randomness that reduces the diversity of sampled responses and thereby constrains subsequent policy optimization.

The behavior of DAPO and DCPO is model-dependent. On the Instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO follow similar declining trajectories. On the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B), however, the trends diverge markedly: DAPO sustains the highest entropy, trending clearly upward but with large fluctuations. On Qwen2.5-Math-7B, GSPO’s entropy closely tracks that of DCPO, and their average metric performance is likewise comparable; on Qwen2.5-14B, by contrast, GSPO’s performance follows a fluctuating downward trend that yields poor results. DCPO instead settles at a moderate entropy level, with a substantially smoother curve confined to a bounded band between the low-entropy regime of GRPO and the high-variance regime of DAPO.

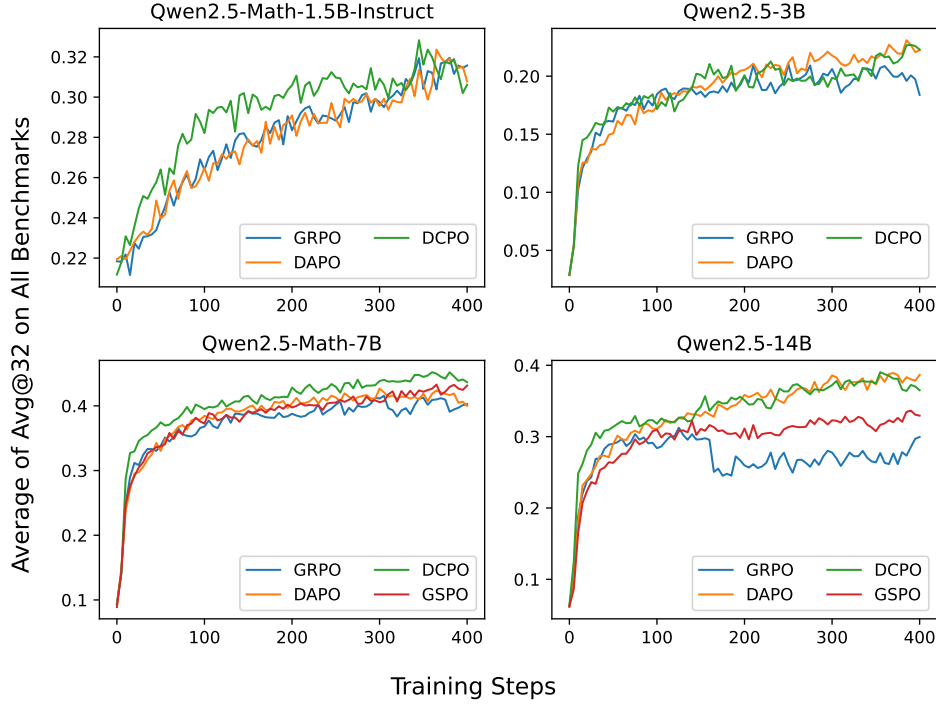


Figure 6: Avg@32 performance over training across the AMC23, AIME24, and AIME25 benchmarks, for each model and method.

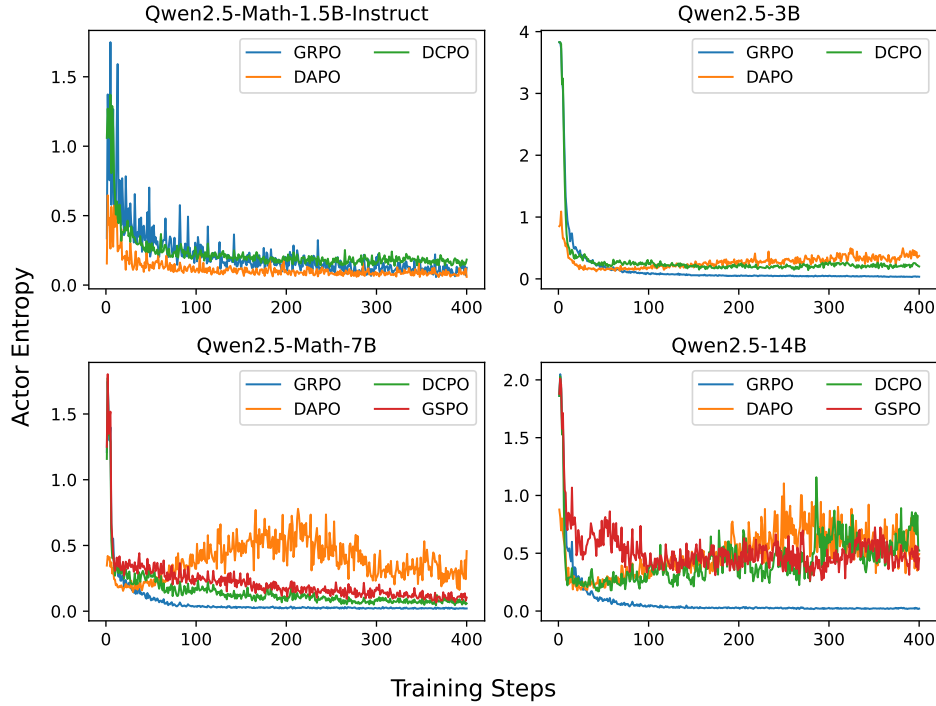


Figure 7: Policy entropy over training for each model and method. DCPO holds entropy within a moderate, stable band between the low-entropy regime of GRPO and the high-variance regime of DAPO.

1080 These observations point to a U-shaped relationship between entropy and training performance:
1081 excessively low entropy starves the learner of exploratory signal and induces premature convergence,
1082 whereas excessively high or unstable entropy undermines training stability. By holding entropy
1083 within a moderate, well-balanced range, DCPO simultaneously promotes convergence and sufficient
1084 exploration—a benefit that grows more pronounced with model capacity. This entropy-regulating
1085 behavior is therefore a plausible mechanism underlying DCPO’s superior reasoning performance and
1086 its more robust policy learning across models of varying sizes.

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